

Parametric Equations & Modeling

ANSWER KEY



Evaluating parametric equations

- 1 Given $x(t) = t^2 - 1$ and $y(t) = 2t + 3$, find the point on the curve at $t = 2$.

$$x(2) = 4 - 1 = 3. \quad y(2) = 4 + 3 = 7. \quad \text{Point: } (3, 7).$$

- 2 Given $x(t) = 3\cos t$ and $y(t) = 3\sin t$, find the point at $t = \frac{\pi}{3}$, and identify the shape traced by the curve.

$$x\left(\frac{\pi}{3}\right) = 3\cos\left(\frac{\pi}{3}\right) = 3\left(\frac{1}{2}\right) = \frac{3}{2}. \quad y\left(\frac{\pi}{3}\right) = 3\sin\left(\frac{\pi}{3}\right) = 3\left(\frac{\sqrt{3}}{2}\right) = \frac{3\sqrt{3}}{2}. \quad \text{Since}$$
$$x^2 + y^2 = 9\cos^2 t + 9\sin^2 t = 9, \text{ the curve traces a circle of radius } 3 \text{ centered at the origin.}$$

Eliminating the parameter

- 3 Eliminate the parameter from $x(t) = t + 1$, $y(t) = t^2 - 2$ to find the rectangular equation, and identify the curve.

From $x = t + 1$: $t = x - 1$. Substitute: $y = (x - 1)^2 - 2 = x^2 - 2x + 1 - 2 = x^2 - 2x - 1$. This is a parabola.

- 4 Eliminate the parameter from $x(t) = 2t - 3$, $y(t) = 4t + 1$ to find the rectangular equation.

From $x = 2t - 3$: $t = \frac{x + 3}{2}$. Substitute: $y = 4\left(\frac{x + 3}{2}\right) + 1 = 2(x + 3) + 1 = 2x + 7$. This is a line: $y = 2x + 7$.

Visualizing a parametric curve's components

- 5 The graph shows the two component functions $x(t) = t^2$ (upper curve) and $y(t) = 2t$ (lower, straight line) for the parametric curve $x(t) = t^2$, $y(t) = 2t$, plotted separately against t on the horizontal axis.

a Eliminate the parameter t to find the rectangular equation relating x and y for this curve. From $y = 2t$: $t = \frac{y}{2}$. Substituting into $x = t^2$: $x = \left(\frac{y}{2}\right)^2 = \frac{y^2}{4}$, so $y^2 = 4x$.

b Use your rectangular equation to describe the shape of the actual parametric curve (not the component graphs shown). $y^2 = 4x$ is a sideways parabola opening to the right, with vertex at the origin — very different from the two component graphs shown, which are just x and y plotted individually against the parameter t .

Parametric equations for lines and conics

- 6 Write parametric equations for the line segment from $(1, 2)$ to $(5, 10)$, using $t \in [0, 1]$.

$x(t) = 1 + 4t$, $y(t) = 2 + 8t$ for $0 \leq t \leq 1$ (since the change in x is $5 - 1 = 4$ and the change in y is $10 - 2 = 8$). Check: at $t = 0$, $(1, 2)$; at $t = 1$, $(5, 10)$ ✓.

- 7 Write parametric equations for an ellipse with semi-major axis 5 (horizontal) and semi-minor axis 3 (vertical), centered at the origin.

$x(t) = 5\cos t$, $y(t) = 3\sin t$, for $0 \leq t < 2\pi$. Check: $\left(\frac{x}{5}\right)^2 + \left(\frac{y}{3}\right)^2 = \cos^2 t + \sin^2 t = 1$, matching the standard ellipse equation.

Modeling projectile motion parametrically

- 8 A ball is launched with initial speed 40 m/s at an angle of 30° above horizontal. Its position is modeled by $x(t) = 40\cos(30^\circ)t$ and $y(t) = 40\sin(30^\circ)t - 4.9t^2$. Find the ball's position at $t = 2$ seconds, to one decimal place.

$$x(2) = 40\cos(30^\circ)(2) \approx 40(0.866)(2) \approx 69.3 \text{ m.}$$

$$y(2) = 40\sin(30^\circ)(2) - 4.9(4) = 40(0.5)(2) - 19.6 = 40 - 19.6 = 20.4 \text{ m. Position: approximately } (69.3, 20.4).$$

- 9 For the same projectile, find the time at which the ball lands (returns to $y = 0$).

Set $y(t) = 0$: $20t - 4.9t^2 = 0$, so $t(20 - 4.9t) = 0$, giving $t = 0$ or $t = \frac{20}{4.9} \approx 4.08$ seconds. The ball lands at $t \approx 4.08$ s.

Comparing parametric speed and direction

- 10 Two objects follow the same path $y = x^2$, but object A uses $x(t) = t$, $y(t) = t^2$ while object B uses $x(t) = 2t$, $y(t) = 4t^2$. Explain why they trace the same curve but do not move through it at the same rate.

Both satisfy $y = x^2$ (for B: $y = 4t^2 = (2t)^2 = x^2$ ✓), so they trace identical curves. But object B's x -coordinate changes twice as fast per unit of t as object A's, meaning B moves through the same path twice as fast — the parametrization encodes speed and direction of travel, which the rectangular equation $y = x^2$ alone does not capture.

Restricting the parameter's domain

- 11 The parametric equations $x(t) = \cos t$, $y(t) = \sin t$ trace the full unit circle for $t \in [0, 2\pi)$. Describe the portion of the circle traced when the domain is restricted to $t \in \left[0, \frac{\pi}{2}\right]$.

Restricting to $t \in \left[0, \frac{\pi}{2}\right]$ traces only the arc from $(1, 0)$ (at $t = 0$) to $(0, 1)$ (at $t = \frac{\pi}{2}$) — the quarter of the circle lying in Quadrant I.

- 12 Explain why restricting t in $x(t) = t^2$, $y(t) = t$ to $t \geq 0$ changes the rectangular relation from $x = y^2$ (a full sideways parabola) to only its upper half.

Eliminating the parameter gives $x = y^2$ for all t , but since $y = t$ directly, restricting $t \geq 0$ forces $y \geq 0$ as well — so only the upper half of the sideways parabola (where $y \geq 0$) is actually traced, even though the unrestricted equation $x = y^2$ would include the lower half too.

Parametric equations for circular motion with a phase shift

- 13 A point moves along a circle of radius 4 according to $x(t) = 4\cos\left(t + \frac{\pi}{4}\right)$, $y(t) = 4\sin\left(t + \frac{\pi}{4}\right)$. Find its position at $t = 0$, and explain the effect of the $\frac{\pi}{4}$ phase shift compared to $x(t) = 4\cos t$, $y(t) = 4\sin t$.

At $t = 0$: $x(0) = 4\cos\left(\frac{\pi}{4}\right) = 4\left(\frac{\sqrt{2}}{2}\right) = 2\sqrt{2}$, $y(0) = 4\sin\left(\frac{\pi}{4}\right) = 2\sqrt{2}$. Without the phase shift, the point would start at $(4, 0)$ when $t = 0$; the added $\frac{\pi}{4}$ advances the starting position by an eighth of a full revolution counterclockwise, to $(2\sqrt{2}, 2\sqrt{2})$, while the circle traced and the speed of travel remain unchanged.

Multiple choice — easy 15

- 14 Given $x(t) = -3t - 2$ and $y(t) = -5t$, find the point on the curve at $t = -2$.

- a. $(10, 4)$
- b. $(4, 10)$ ✓
- c. $(4, 11)$
- d. $(5, 10)$

$x(-2) = -3(-2) - 2 = 4$. $y(-2) = -5(-2) = 10$. Point: $(4, 10)$.

- 15 Given $x(t) = -2t - 5$ and $y(t) = 2t + 6$, find the point on the curve at $t = -5$.

- a. $(6, -4)$
- b. $(-4, 5)$
- c. $(5, -3)$
- d. $(5, -4)$ ✓

$x(-5) = -2(-5) - 5 = 5$. $y(-5) = 2(-5) + 6 = -4$. Point: $(5, -4)$.

16 Given $x(t) = -5t - 5$ and $y(t) = 2t - 3$, find the point on the curve at $t = 2$.

- a. $(-15, 2)$
- b. $(-14, 1)$
- c. $(-15, 1)$ ✓
- d. $(1, -15)$

$$x(2) = -5(2) - 5 = -15. \quad y(2) = 2(2) - 3 = 1. \quad \text{Point: } (-15, 1).$$

17 Given $x(t) = -6t - 3$ and $y(t) = -4t - 5$, find the point on the curve at $t = 3$.

- a. $(-20, -17)$
- b. $(-21, -17)$ ✓
- c. $(-21, -16)$
- d. $(-17, -21)$

$$x(3) = -6(3) - 3 = -21. \quad y(3) = -4(3) - 5 = -17. \quad \text{Point: } (-21, -17).$$

18 Given $x(t) = -2t + 9$ and $y(t) = 3t + 5$, find the point on the curve at $t = 0$.

- a. $(9, 5)$ ✓
- b. $(9, 6)$
- c. $(10, 5)$
- d. $(5, 9)$

$$x(0) = -2(0) + 9 = 9. \quad y(0) = 3(0) + 5 = 5. \quad \text{Point: } (9, 5).$$

19 Given $x(t) = 2t - 5$ and $y(t) = -2t + 2$, find the point on the curve at $t = 1$.

- a. $(-3, 0)$ ✓
- b. $(-3, 1)$
- c. $(-2, 0)$
- d. $(0, -3)$

$$x(1) = 2(1) - 5 = -3. \quad y(1) = -2(1) + 2 = 0. \quad \text{Point: } (-3, 0).$$

20 Given $x(t) = -3t + 4$ and $y(t) = -3t + 4$, find the point on the curve at $t = -3$.

a. $(13, 13)$ ✓

b. $(11, 15)$

c. $(14, 13)$

d. $(13, 14)$

$x(-3) = -3(-3) + 4 = 13$. $y(-3) = -3(-3) + 4 = 13$. Point: $(13, 13)$.

21 Given $x(t) = -t + 5$ and $y(t) = -5t - 7$, find the point on the curve at $t = 0$.

a. $(-7, 5)$

b. $(5, -6)$

c. $(5, -7)$ ✓

d. $(6, -7)$

$x(0) = -1(0) + 5 = 5$. $y(0) = -5(0) - 7 = -7$. Point: $(5, -7)$.

22 Given $x(t) = 4t - 7$ and $y(t) = t - 7$, find the point on the curve at $t = -5$.

a. $(-27, -12)$ ✓

b. $(-26, -12)$

c. $(-27, -11)$

d. $(-12, -27)$

$x(-5) = 4(-5) - 7 = -27$. $y(-5) = 1(-5) - 7 = -12$. Point: $(-27, -12)$.

23 Given $x(t) = -t - 3$ and $y(t) = 5t - 6$, find the point on the curve at $t = 0$.

- a. $(-2, -6)$
- b. $(-3, -6)$ ✓
- c. $(-6, -3)$
- d. $(-3, -5)$

$$x(0) = -1(0) - 3 = -3. y(0) = 5(0) - 6 = -6. \text{ Point: } (-3, -6).$$

24 Given $x(t) = 6t + 9$ and $y(t) = 2t + 5$, find the point on the curve at $t = -2$.

- a. $(-2, 1)$
- b. $(-3, 1)$ ✓
- c. $(-3, 2)$
- d. $(1, -3)$

$$x(-2) = 6(-2) + 9 = -3. y(-2) = 2(-2) + 5 = 1. \text{ Point: } (-3, 1).$$

25 Given $x(t) = 2t + 4$ and $y(t) = -6t - 7$, find the point on the curve at $t = 3$.

- a. $(-25, 10)$
- b. $(10, -25)$ ✓
- c. $(11, -25)$
- d. $(10, -24)$

$$x(3) = 2(3) + 4 = 10. y(3) = -6(3) - 7 = -25. \text{ Point: } (10, -25).$$

26 Given $x(t) = 4t + 4$ and $y(t) = t + 9$, find the point on the curve at $t = -6$.

- a. $(-20, 4)$
- b. $(-19, 3)$
- c. $(3, -20)$
- d. $(-20, 3)$ ✓

$$x(-6) = 4(-6) + 4 = -20. y(-6) = 1(-6) + 9 = 3. \text{ Point: } (-20, 3).$$

27 Given $x(t) = 3t - 5$ and $y(t) = t + 1$, find the point on the curve at $t = -6$.

- a. $(-22, -5)$
- b. $(-23, -5)$ ✓
- c. $(-23, -4)$
- d. $(-5, -23)$

$x(-6) = 3(-6) - 5 = -23$. $y(-6) = 1(-6) + 1 = -5$. Point: $(-23, -5)$.

28 Given $x(t) = 6t - 4$ and $y(t) = 5t - 3$, find the point on the curve at $t = 5$.

- a. $(27, 22)$
- b. $(22, 26)$
- c. $(26, 22)$ ✓
- d. $(26, 23)$

$x(5) = 6(5) - 4 = 26$. $y(5) = 5(5) - 3 = 22$. Point: $(26, 22)$.

Multiple choice — medium 15

29 Eliminate the parameter from $x(t) = t - 2$ and $y(t) = -3t + 4$ to find the rectangular equation for y in terms of x .

- a. $y = -3x - 1$
- b. $y = -3x + 4$
- c. $y = 3x - 2$
- d. $y = -3x - 2$ ✓

From $x = t - 2$: $t = x + 2$. Substitute: $y = -3(x + 2) + 4 = -3x - 2$.

- 30 Eliminate the parameter from $x(t) = t - 9$ and $y(t) = -7t + 7$ to find the rectangular equation for y in terms of x .

- a. $y = 7x - 56$
- b. $y = -7x - 55$
- c. $y = -7x + 7$
- d. $y = -7x - 56$ ✓

From $x = t - 9$: $t = x + 9$. Substitute: $y = -7(x + 9) + 7 = -7x - 56$.

- 31 Eliminate the parameter from $x(t) = t - 8$ and $y(t) = 7t - 9$ to find the rectangular equation for y in terms of x .

- a. $y = 7x - 9$
- b. $y = -7x + 47$
- c. $y = 7x + 47$ ✓
- d. $y = 7x + 48$

From $x = t - 8$: $t = x + 8$. Substitute: $y = 7(x + 8) - 9 = 7x + 47$.

- 32 Eliminate the parameter from $x(t) = t + 5$ and $y(t) = 7t - 2$ to find the rectangular equation for y in terms of x .

- a. $y = 7x - 36$
- b. $y = 7x - 37$ ✓
- c. $y = 7x - 2$
- d. $y = -7x - 37$

From $x = t + 5$: $t = x - 5$. Substitute: $y = 7(x - 5) - 2 = 7x - 37$.

- 33** Eliminate the parameter from $x(t) = t - 6$ and $y(t) = -5t - 4$ to find the rectangular equation for y in terms of x .

a. $y = 5x - 34$

b. $y = -5x - 34$ ✓

c. $y = -5x - 33$

d. $y = -5x - 4$

From $x = t - 6$: $t = x + 6$. Substitute: $y = -5(x + 6) - 4 = -5x - 34$.

- 34** Eliminate the parameter from $x(t) = t - 1$ and $y(t) = 8t - 5$ to find the rectangular equation for y in terms of x .

a. $y = 8x - 5$

b. $y = -8x + 3$

c. $y = 8x + 3$ ✓

d. $y = 8x + 4$

From $x = t - 1$: $t = x + 1$. Substitute: $y = 8(x + 1) - 5 = 8x + 3$.

- 35** Eliminate the parameter from $x(t) = t + 2$ and $y(t) = 6t - 1$ to find the rectangular equation for y in terms of x .

a. $y = 6x - 1$

b. $y = -6x - 13$

c. $y = 6x - 12$

d. $y = 6x - 13$ ✓

From $x = t + 2$: $t = x - 2$. Substitute: $y = 6(x - 2) - 1 = 6x - 13$.

- 36 Eliminate the parameter from $x(t) = t - 5$ and $y(t) = t - 7$ to find the rectangular equation for y in terms of x .

a. $y = -x - 2$

b. $y = x - 7$

c. $y = x - 2$ ✓

d. $y = x - 1$

From $x = t - 5$: $t = x + 5$. Substitute: $y = 1(x + 5) - 7 = x - 2$.

- 37 Eliminate the parameter from $x(t) = t + 3$ and $y(t) = -3t - 3$ to find the rectangular equation for y in terms of x .

a. $y = 3x + 6$

b. $y = -3x + 6$ ✓

c. $y = -3x - 3$

d. $y = -3x + 7$

From $x = t + 3$: $t = x - 3$. Substitute: $y = -3(x - 3) - 3 = -3x + 6$.

- 38 Eliminate the parameter from $x(t) = t + 6$ and $y(t) = t + 4$ to find the rectangular equation for y in terms of x .

a. $y = x - 2$ ✓

b. $y = -x - 2$

c. $y = x - 1$

d. $y = x + 4$

From $x = t + 6$: $t = x - 6$. Substitute: $y = 1(x - 6) + 4 = x - 2$.

- 39 Eliminate the parameter from $x(t) = t - 6$ and $y(t) = -3t - 3$ to find the rectangular equation for y in terms of x .

- a. $y = -3x - 20$
- b. $y = 3x - 21$
- c. $y = -3x - 21$ ✓
- d. $y = -3x - 3$

From $x = t - 6$: $t = x + 6$. Substitute: $y = -3(x + 6) - 3 = -3x - 21$.

- 40 Eliminate the parameter from $x(t) = t + 9$ and $y(t) = 3t + 4$ to find the rectangular equation for y in terms of x .

- a. $y = 3x - 23$ ✓
- b. $y = 3x - 22$
- c. $y = 3x + 4$
- d. $y = -3x - 23$

From $x = t + 9$: $t = x - 9$. Substitute: $y = 3(x - 9) + 4 = 3x - 23$.

- 41 Eliminate the parameter from $x(t) = t$ and $y(t) = 8t - 9$ to find the rectangular equation for y in terms of x .

- a. $y = 8x - 9$ ✓
- b. $y = -8x - 9$
- c. $y = 8x - 7$
- d. $y = 8x - 8$

From $x = t$: $t = x$. Substitute: $y = 8(x) - 9 = 8x - 9$.

- 42 Eliminate the parameter from $x(t) = t - 2$ and $y(t) = -8t + 3$ to find the rectangular equation for y in terms of x .

- a. $y = -8x + 3$
- b. $y = -8x - 12$
- c. $y = 8x - 13$
- d. $y = -8x - 13$ ✓

From $x = t - 2$: $t = x + 2$. Substitute: $y = -8(x + 2) + 3 = -8x - 13$.

- 43 Eliminate the parameter from $x(t) = t - 2$ and $y(t) = -t - 5$ to find the rectangular equation for y in terms of x .

- a. $y = -x - 7$ ✓
- b. $y = -x - 6$
- c. $y = x - 7$
- d. $y = -x - 5$

From $x = t - 2$: $t = x + 2$. Substitute: $y = -1(x + 2) - 5 = -x - 7$.

Multiple choice — hard 15

- 44 Given $x(t) = t^2 + 6$ and $y(t) = -2t + 8$, find the point on the curve at $t = 0$.

- a. (8, 8)
- b. (6, 8) ✓
- c. (8, 6)
- d. (6, 10)

$x(0) = (0)^2 + 6 = 6$. $y(0) = -2(0) + 8 = 8$. Point: (6, 8).

45 Given $x(t) = t^2 + 2$ and $y(t) = -2t + 7$, find the point on the curve at $t = -1$.

a. (3, 9) ✓

b. (9, 3)

c. (5, 9)

d. (3, 11)

$$x(-1) = (-1)^2 + 2 = 3. \quad y(-1) = -2(-1) + 7 = 9. \quad \text{Point: } (3, 9).$$

46 Given $x(t) = t^2 + 4$ and $y(t) = 2t - 2$, find the point on the curve at $t = 4$.

a. (20, 8)

b. (22, 6)

c. (6, 20)

d. (20, 6) ✓

$$x(4) = (4)^2 + 4 = 20. \quad y(4) = 2(4) - 2 = 6. \quad \text{Point: } (20, 6).$$

47 Given $x(t) = t^2 - 9$ and $y(t) = 4t + 3$, find the point on the curve at $t = -1$.

a. (-8, 1)

b. (-1, -8)

c. (-6, -1)

d. (-8, -1) ✓

$$x(-1) = (-1)^2 - 9 = -8. \quad y(-1) = 4(-1) + 3 = -1. \quad \text{Point: } (-8, -1).$$

48 Given $x(t) = t^2 + 9$ and $y(t) = -t - 7$, find the point on the curve at $t = 2$.

a. (13, -9) ✓

b. (-9, 13)

c. (13, -7)

d. (15, -9)

$$x(2) = (2)^2 + 9 = 13. \quad y(2) = -1(2) - 7 = -9. \quad \text{Point: } (13, -9).$$

49 A curve is given by $x(t) = 4\cos t$, $y(t) = 4\sin t$. Find the exact point on the curve at $t = \frac{11\pi}{6}$ (330°).

- a. $(2\sqrt{3}, 2)$
- b. $(2\sqrt{3}, -2)$ ✓
- c. $(-2, 2\sqrt{3})$
- d. $(-2\sqrt{3}, -2)$

$$x = 4\cos\left(\frac{11\pi}{6}\right) = 2\sqrt{3}. y = 4\sin\left(\frac{11\pi}{6}\right) = -2. \text{ Point: } (2\sqrt{3}, -2).$$

50 A curve is given by $x(t) = 8\cos t$, $y(t) = 8\sin t$. Find the exact point on the curve at $t = \frac{5\pi}{4}$ (225°).

- a. $(-4\sqrt{2}, -4\sqrt{2})$ ✓
- b. $(4\sqrt{2}, -4\sqrt{2})$
- c. $(0, -8)$
- d. $(-4\sqrt{2}, 4\sqrt{2})$

$$x = 8\cos\left(\frac{5\pi}{4}\right) = -4\sqrt{2}. y = 8\sin\left(\frac{5\pi}{4}\right) = -4\sqrt{2}. \text{ Point: } (-4\sqrt{2}, -4\sqrt{2}).$$

51 A curve is given by $x(t) = 8\cos t$, $y(t) = 8\sin t$. Find the exact point on the curve at $t = \frac{3\pi}{2}$ (270°).

- a. $(0, -8)$ ✓
- b. $(-8, 0)$
- c. $(-4\sqrt{3}, -4)$
- d. $(0, 8)$

$$x = 8\cos\left(\frac{3\pi}{2}\right) = 0. y = 8\sin\left(\frac{3\pi}{2}\right) = -8. \text{ Point: } (0, -8).$$

52 A curve is given by $x(t) = 4\cos t$, $y(t) = 4\sin t$. Find the exact point on the curve at $t = \frac{4\pi}{3}$ (240°).

a. $(-2, 2\sqrt{3})$

b. $(2, -2\sqrt{3})$

c. $(-2\sqrt{3}, -2)$

d. $(-2, -2\sqrt{3})$ ✓

$$x = 4\cos\left(\frac{4\pi}{3}\right) = -2. \quad y = 4\sin\left(\frac{4\pi}{3}\right) = -2\sqrt{3}. \quad \text{Point: } (-2, -2\sqrt{3}).$$

53 A curve is given by $x(t) = 4\cos t$, $y(t) = 4\sin t$. Find the exact point on the curve at $t = \frac{7\pi}{4}$ (315°).

a. $(2\sqrt{2}, 2\sqrt{2})$

b. $(-2\sqrt{2}, 2\sqrt{2})$

c. $(-2\sqrt{2}, -2\sqrt{2})$

d. $(2\sqrt{2}, -2\sqrt{2})$ ✓

$$x = 4\cos\left(\frac{7\pi}{4}\right) = 2\sqrt{2}. \quad y = 4\sin\left(\frac{7\pi}{4}\right) = -2\sqrt{2}. \quad \text{Point: } (2\sqrt{2}, -2\sqrt{2}).$$

54 A projectile is launched with initial speed 34 m/s at 30° above horizontal, so $y(t) = 34\sin(30^\circ)t - 4.9t^2$. Find $y(2)$, to one decimal place.

a. 12.4

b. 19.4

c. 16.4

d. 14.4 ✓

$$y(2) = 34\sin(30^\circ)(2) - 4.9(2)^2 \approx 34(0.5)(2) - 4.9(4) \approx 14.4 \text{ m.}$$

- 55 A projectile is launched with initial speed 49 m/s at 60° above horizontal, so $y(t) = 49\sin(60^\circ)t - 4.9t^2$. Find $y(2)$, to one decimal place.

a. 65.3 ✓

b. 70.3

c. 67.3

d. 63.3

$$y(2) = 49\sin(60^\circ)(2) - 4.9(2)^2 \approx 49(0.86603)(2) - 4.9(4) \approx 65.3 \text{ m.}$$

- 56 A projectile is launched with initial speed 45 m/s at 30° above horizontal, so $y(t) = 45\sin(30^\circ)t - 4.9t^2$. Find $y(2)$, to one decimal place.

a. 23.4

b. 25.4 ✓

c. 30.4

d. 27.4

$$y(2) = 45\sin(30^\circ)(2) - 4.9(2)^2 \approx 45(0.5)(2) - 4.9(4) \approx 25.4 \text{ m.}$$

- 57 A projectile is launched with initial speed 60 m/s at 30° above horizontal, so $y(t) = 60\sin(30^\circ)t - 4.9t^2$. Find $y(1)$, to one decimal place.

a. 27.1

b. 25.1 ✓

c. 23.1

d. 30.1

$$y(1) = 60\sin(30^\circ)(1) - 4.9(1)^2 \approx 60(0.5)(1) - 4.9(1) \approx 25.1 \text{ m.}$$

58 A projectile is launched with initial speed 59 m/s at 60° above horizontal, so $y(t) = 59\sin(60^\circ)t - 4.9t^2$. Find $y(4)$, to one decimal place.

a. 128.0

b. 126.0 ✓

c. 124.0

d. 131.0

$$y(4) = 59\sin(60^\circ)(4) - 4.9(4)^2 \approx 59(0.86603)(4) - 4.9(16) \approx 126.0 \text{ m.}$$